

Dr. Tanveer Karim

DUNLAP POSTDOCTORAL FELLOW · UNIVERSITY OF TORONTO

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Research Interests

My research focuses on the intersection of high redshift large-scale structure surveys and Bayesian inference. Methodologically, I work on designing robust frameworks that can combine multiple surveys to provide unbiased constraints of fundamental physics models, especially with a focus on dark energy and galaxy physics.

Professional Experiences

- 2025-Pres **Dunlap Postdoctoral Fellow**, University of Toronto
- 2023-2025 **Arts & Sciences Postdoctoral Fellow**, University of Toronto
- Sum 2023 **Postdoctoral Fellow**, Harvard University

Education

Harvard University

Cambridge MA, USA

PHD IN ASTRONOMY & ASTROPHYSICS

2017 - 2023

- *Precision Cosmology with Emission-Line Galaxies*. Advisor: Daniel Eisenstein

University of Rochester

Rochester NY, USA

BS IN PHYSICS & ASTRONOMY

2012 - 2017

- Magna Cum Laude, Highest Distinction
- Astronomy Thesis: *Kinematic Explorations of T-Tauri Stars and the Fermi Bubbles*. Advisors: Andrew Fox and Eric Blackman
- Russian Thesis: *From Circassia to Chechnya: Depiction of the Caucasus through the lens of Tolstoy*. Advisor: John Givens

Selected Awards, Fellowships & Grants

2026	Lindau Nobel Laureate Meetings , Krupp Fellow	EUR 5,000
2025	Catalyst Postdoc Alliance , LSST-Discovery Alliance	
2025	RCSA Fellowship , Research Corporation for Science Advancement	
2025	Conference Grant for Cosmology From Lyman-Break Galaxies , Dunlap Institute & Canadian Institute of Theoretical Astrophysics	CAD 6,500
2024	Excellence Award for Summer Interns , University of Toronto	CAD 7,500
2023	Arts & Sciences Postdoctoral Fellowship , University of Toronto	CAD 96,000
	Kavli Fellow , National Academy of Sciences	
	Barbara Bell Dissertation Fellowship , Harvard University	USD 25,000
2022	Ford Foundation Dissertation Fellowship , Honorable Mention, Ford Foundation	
2019	National Science Foundation Graduate Fellowship , National Science Foundation	USD 147,000
2016	Goldwater Scholar , The Barry Goldwater Scholarship Foundation	USD 7,500
2015	Take Five Scholar , University of Rochester	USD 40,000

Collaboration Memberships

- 2018-Pres **Dark Energy Spectroscopic Instrument Collaboration**, Builder
 - 2023-Pres **Dark Energy Science Collaboration**, Full Member
 - 2025-Pres **Center for Astronomy, Space Science and Astrophysics, Independent University**, Associate Member
- Bangladesh

Leadership & Service

2025-Pres	DESI Collaboration , Photometric Redshift Topical Team Co-Lead	
2024-Pres	Dark Energy Science Collaboration , Lyman-Break Galaxies Topical Team Co-Lead	
2025-Pres	Dark Energy Science Collaboration , Paper Review Committee	
2024-2025	DESI Collaboration , Education, Publicity and Outreach Committee	
2025	Cosmology from Lyman-Break Galaxies Workshop , Main Organizer	Toronto
2024	Dark Energy Science Collaboration , Collaboration Meeting Meetings Contact Person	Urbana
2024	Dark Energy Science Collaboration , Collaboration Meeting Scientific Organizing Committee	Zurich
2023	Department of Astronomy Graduate Admissions Committee, Harvard University , Committee Member	
2021-2022	Diversity, Equity and Inclusion Committee, DESI , Committee Member	
2018-2019	Equity & Inclusion Journal Club, Harvard University , Organizer	Cambridge

Mentoring

Pres	Amani Hussien , Undergraduate, Constraining Modified Gravity using Marked Field and CMB lensing, U Toronto
Pres	Alex Brisebois , Graduate Student, Constraining Redshift-Evolving σ_8 in LSST Era, U Toronto
Pres	Adrita Khan , Graduate student, Modified Gravity Signatures in Galaxy Clustering and CMB Lensing, Independent University, Bangladesh
Pres	Ivan Kraskov , Undergrad, Simulation of Intergalactic Medium Attenuation, U Toronto
2025-2026	Malaika Malik , Undergrad, Robust Emulator Architecture Design for Cosmological Inference, U Toronto
2024-2025	Harmandeep Gill , Undergrad, Simulation of Intergalactic Medium Attenuation, U Toronto
2024-2025	Avery Brown , Undergrad, Cross-matching Lyman-Break Galaxies Catalogs, U Toronto
2024-2025	Everest Guo , Undergrad, Photometric Redshift of Lyman-Break Galaxies, U Toronto
2024	Seo Won Yi , Undergrad, Bayesian Model Selection Metrics, U Toronto

Teaching Experience

2024-Pres	Introduction to Astrophysical Research , Research Advisor, University of Toronto
Spr 2022	Life as a Planetary Phenomenon , Teaching Fellow & Guest Lecturer, Harvard University
Spr 2020	Prediction: The Past & The Present of the Future , Teaching Fellow, Harvard University
Sum 2019	Public Speaking for Scientists , Instructor, Bancker Institute, Harvard University
Sum 2019	Celestial Coordinate Systems , Instructor, Bancker Institute, Harvard University
Spr 2019	Life as a Planetary Phenomenon , Teaching Fellow, Harvard University
Spr 2016	Relativity, Black Holes & the Big Bang , Teaching Assistant, University of Rochester
Spr 2015	Mechanics for Engineers (Flipped Classroom) , Teaching Assistant, University of Rochester
Spr 2014	Cosmic Origins of Life , Teaching Assistant, University of Rochester
Fall 2013	General Physics I , Teaching Assistant, University of Rochester

Publications

FIRST AUTHOR PUBLICATIONS (130 CITATIONS)

- Karim, T.**, Rezaie M., Singh S., Eisenstein, D. et al., Measuring σ_8 using DESI Legacy Imaging Surveys Emission-Line Galaxies and Planck CMB Lensing and the Impact of Dust on Parameter Inference. arXiv: 2408.15909. JCAP, 02, 045
- Karim, T.**, Rezaie M., Singh S. & Eisenstein, D. 2023. On the Impact of Observational Systematics on Cosmological Parameter Inference. MNRAS, 525, 311 - 324
- Karim, T.**, Lee, J., Eisenstein, D., Burtin, E. et al., 2020. Validation of Emission-Line Galaxies Target Selection Algorithms for the Dark Energy Spectroscopic Survey Using the MMT Binospec, MNRAS, 497, 4587
- Karim, T.**, Fox A., Jenkins, E. B., Bordoloi, R. et al., 2018. Probing the Southern Fermi Bubble in Ultraviolet Absorption Using Distant AGNs, ApJ, 860, 2, 98
- Karim, T.** & Mamajek, E., 2017. Revised Geometric Estimates of the North Galactic Pole and the Sun's Height Above the Galactic Midplane, MNRAS, 465, 472
- Karim, T.**, Stassun, K. G., Briceño, C., Vivas, A. K. et al., 2016. The Rotation Period Distributions of 4 – 10 Myr T Tauri Stars in Orion OB1: New Constraints on Pre-main-sequence Angular Momentum Evolution, AJ, 152, 198

CO-AUTHORED PUBLICATIONS(408 CITATIONS)

- Finkbeiner, D. P., Speagle, J. S. & **Karim, T.**, 2025. Data-Driven Stellar Spectral Modelling with GSPICE, **Submitted**.
- Crenshaw, J. F., Leistedt, B., Graham, M. L., Payerne, C. et al., 2025. Quantifying the Impact of LSST *u*-band Survey Strategy on Photometric Redshift Estimation and the Detection of Lyman-break Galaxies, **Submitted**.
- Ross, A. J. et al., 2025. The Construction of Large-scale Structure Catalogs for the Dark Energy Spectroscopic Instrument, JCAP, 01, 125
- Raichoor, A., Moutakas, J., Newman, J. A., **Karim, T.** et al., 2023. Target Selection and Validation of DESI Emission Line Galaxies, AJ, 165, 3, 126
- Myers, A., Moutakas, J., Bailey, S., Weaver, B. et al., 2023. The Target-selection Pipeline for the Dark Energy Spectroscopic Instrument, AJ, 165, 2, 50
- Han, J., Dey, A., Price-Whelan, A., Najita, J. et al., 2023. NANCY: Next-generation All-sky Near-infrared Community survey, Roman Core Community Survey White Paper
- Ashley, T., Fox, A. J., Cashman, F. H., Lockman, F. J. et al., 2022. Diverse metallicities of Fermi bubble clouds indicate dual origins in the disk and halo, Nature Astronomy, 6, 968 – 975
- Raichoor, A., Eisenstein, D., **Karim, T.**, Newman, J. A. et al., 2020. Preliminary Target Selection for the DESI Emission Line Galaxy (ELG) Sample, RNAAS, 4, 10, 180
- Ashley, T., Fox, A. J., Jenkins, E. B., Wakker, B. P. et al., 2020. Mapping Outflowing Gas in the Fermi Bubbles: A UV Absorption Survey of the Galactic Nuclear Wind, ApJ, 898, 2, 128

COLLABORATION-WIDE PUBLICATIONS (1045 CITATIONS)

- DESI Collaboration et al., 2025. Data Release 1 of the Dark Energy Spectroscopic Instrument, **Submitted** to AJ.
- DESI Collaboration et al., 2024. The Early Data Release of the Dark Energy Spectroscopic Instrument, AJ, 168, 2
- DESI Collaboration et al., 2024. Validation of the Scientific Program for the Dark Energy Spectroscopic Instrument, AJ, 167
- DESI Collaboration et al., 2022. Overview of the Instrumentation for the Dark Energy Spectroscopic Instrument, AJ, 164, 5

Presentations

INVITED SCIENCE TALKS

- July 2026. *Isolating Dark Energy: How to Measure the Universe When Galaxies Get in the Way*. 75th Lindau Nobel Laureate Meetings, Lindau, Germany.
- Dec 2025. *Designing Emission-Line Galaxies Lightcone Mocks for CMB Lensing Cross-Correlation*. DESI Collaboration Meeting, AZ, USA.
- Nov 2025. *Constraining Cosmology and Galaxy Physics using Emission-Line Galaxies and CMB secondary anisotropies*. Theoretical Astronomy and Cosmology Seminar, University of Arizona, AZ, USA.
- Oct 2025. *From Star-Forming Galaxies to the Cosmos: A Unified Vision for Wide-Field Surveys*. Department of Physics Colloquium, Illinois State University, IL, USA.
- Sept 2025. *Constraining Cosmology and Galaxy Physics using Emission-Line Galaxies and CMB secondary anisotropies*. Astronomy - Particle Physics - Experimental Physics - Cosmology Seminar, Kavli IPMU, University of Tokyo, Japan.
- Sept 2025. *Survey Systematics for Galaxy-CMB Cross-Correlation Analysis*. Beyond-2-Point Statistics Meet Survey Systematics, Kavli IPMU, University of Tokyo, Japan.
- Aug 2025. *Unlocking Galaxy History with Cosmic Infrared Background: A first look at DESI ELG X Planck CIB*. Dunlap Institute Lunch Seminar TASTY, University of Toronto, Canada.
- Feb 2025. *Observational Considerations for Emission-Line Galaxies*. ELG Modeling and Mock Challenge Workshop, DIPC, Donostia, Spain.
- Jan 2025. *Unraveling the Distant Universe with Wide-Field Surveys of Star-Forming Galaxies*. Independent University, Bangladesh Astronomy Colloquium, Dhaka, Bangladesh.
- Fall 2024. *Measuring the Growth of Structure with Emission-Line Galaxies from the DESI Legacy Imaging Surveys and Planck CMB Lensing*. Center for Computational Astrophysics Data Science X Cosmology Seminar, New York, NY USA.
- Fall 2024. *Large-Scale Structures at $z \geq 1$: Opportunities with Tomography and Cross-Correlations*. University of Waterloo AstroSeminar, Waterloo, ON Canada.
- Summer 2024. *Observational Considerations in Spectroscopic Surveys*. ML for Fundamental Physics, Aspen Center for Physics, CO USA.
- Spring 2024. *Imaging Systematics Estimator is Biased in Power Spectrum*. Dark Energy Science Collaboration Seminar.
- Spring 2024. *Illuminating the Dark Universe*. National Academy of Sciences Kavli Frontiers of Science Symposium plenary, CA USA.
- Fall 2023. *Constraining Cosmology with Emission-Line Galaxies*. Yale Cosmology Seminar, CT USA.

INVITED PUBLIC ENGAGEMENT TALKS

- Spring 2025. *Unraveling the Mystery of Dark Energy with 5000 Eyes*. Toronto Astronomy on Tap, Toronto, Canada.
- Summer 2023. *Measuring the Growth of the Universe with Multiple Datasets*. Nantucket, Massachusetts, USA.
- Summer 2022. *Model Minority myth and its impact on policymaking: A case study on Asian-Americans*. European Astronomical Society Early Career Astronomers Session, Valencia, Spain.